CAPSTONE PROJECT PROPOSAL

BioSci Teaching Assistant Application and Selection Management System

Submitting Unit: UC Irvine School of Biological Sciences (BioSci), Graduate Studies and Support

Capstone Term: Fall 2026, 10-week quarter

PROJECT SUMMARY

The UC Irvine School of Biological Sciences seeks to develop a centralized, web-based system supporting the Teaching Assistant application, identification, matching, and selection process.

Students will use an accessible online application to submit their academic information, UCI student ID, student level, weekly availability, teaching experience, relevant skills, and course interests. Authorized BioSci personnel will use the same system to manage course and staffing information, review applicants, identify potential student-course matches, record selection decisions, and export final assignment rosters.

The current process relies heavily on disconnected spreadsheets and email communication. The proposed system will create a consistent source of applicant, course, and selection information while preserving human authority over all recommendations and assignments.

The system will support departmental decision-making but will not independently select applicants or perform formal hiring, appointment, payroll, or other Human Resources functions.

BACKGROUND AND PROBLEM STATEMENT

BioSci's current TA application and selection activities involve information exchanged through multiple spreadsheets and email threads. Principal Investigators, the Vice Chair, Graduate Studies and Support staff, and potentially other departmental stakeholders participate in the process. The exact review and approval structure may vary and will be confirmed during project discovery.

The current approach creates several challenges:

- Applicant information may be incomplete, duplicated, or stored in inconsistent formats.
- Student availability can be difficult to compare with lecture, laboratory, and discussion schedules.
- Relevant teaching, laboratory, research, programming, and technical skills may not be systematically connected to course requirements.

- Student availability may not always be captured or incorporated consistently into selection decisions.
- Recommendations, approvals, and final selections can be difficult to track across stakeholders.
- It can be difficult to determine which TA positions are proposed, filled, or still vacant.
- Stakeholders may not have consistent or appropriately controlled access to current information.
- Administrative staff must spend significant time organizing and reconciling data before it can support decisions.

A centralized application and selection-management system would provide one authoritative source of information for each application cycle while preserving human oversight of all TA assignments.

PROJECT OBJECTIVES

The project's primary objectives are to:

1. Create a student-facing TA application that is easy to access from the BioSci website.
2. Collect student availability, qualifications, interests, experience, and academic information in a consistent format.
3. Create an administrative interface for managing application periods, applicant records, course requirements, and available TA positions.
4. Help authorized users identify potential student-course matches based on availability, skills, experience, interests, and other defined criteria.
5. Allow authorized personnel to record recommendations, selections, assignments, and related statuses in the same system.
6. Provide a clear view of proposed, filled, and unfilled TA positions.
7. Reduce reliance on disconnected spreadsheets and email.
8. Support controlled access for administrators, reviewers, decision-makers, and approved viewers.
9. Produce final assignment rosters and other exportable reports.
10. Provide a maintainable system that BioSci personnel can operate without specialized technical expertise.

PROPOSED SCOPE AND CORE FUNCTIONALITY

Student Application

The student-facing application should collect information that may include:

- Full name
- UCI email address
- UCI student ID
- Academic program or department
- Student level: undergraduate, master's, or PhD
- Weekly availability and known scheduling conflicts
- Courses, disciplines, or subject areas of interest
- Previous TA or teaching experience
- Relevant laboratory, research, programming, or technical skills
- Required training status, when applicable
- Additional qualifications, constraints, or comments
- Required acknowledgments or certifications

The exact applicant eligibility rules and final application fields will be confirmed during project discovery. The system should be flexible enough to support different student populations and future eligibility changes.

Application Period Management

BioSci administrators should be able to:

- Open the student-facing application for a defined period
- Close the application when submissions are no longer being accepted
- Retain administrative access after the student-facing application closes
- Reopen the application if necessary
- Clearly indicate whether applications are open or closed

Course, Section, and Staffing Management

Authorized administrators should be able to create, edit, upload, or otherwise maintain course and staffing information. Records may include:

- Course number and title
- Instructor or responsible faculty member
- Lecture, laboratory, and discussion sections
- Section meeting days and times
- Number of available TA positions
- Required skills
- Preferred skills

- Relevant subject-matter experience
- Appointment percentage or anticipated workload, if applicable
- Additional course-specific requirements or notes

For example, a course could identify proficiency in Python programming as a required or preferred qualification.

The method for entering course information—such as manual entry or structured spreadsheet upload—will be determined during discovery. Integration with another UCI system is not required for the initial minimum viable product.

Applicant Identification and Matching

The administrative interface should allow authorized users to:

- View, search, sort, and filter applications
- Compare student availability with course and section schedules
- Filter applicants by student level, experience, skills, and interests
- Identify applicants who satisfy required course criteria
- View suggested student-course or student-section matches
- Understand which criteria contributed to a suggested match
- Identify possible scheduling conflicts or missing qualifications
- Add internal notes, flags, or review statuses
- Override or disregard system-generated recommendations

Matching recommendations will function as decision-support tools. The system will not independently determine who should receive a TA assignment.

Selection and Assignment Management

The system should allow authorized users to:

- Place an applicant under consideration for a course or section
- Record a recommendation or proposed assignment
- Mark a candidate as selected, alternate, declined, or not selected
- Assign a selected applicant to an available TA position
- Update or remove a proposed assignment
- Track the status of each position and applicant
- Identify possible schedule conflicts or duplicate assignments
- View filled, proposed, and unfilled positions

- Record the authorized user responsible for significant status or assignment changes
- Export a final assignment roster to Excel or CSV

The precise recommendation and approval sequence will be confirmed during discovery. The system should use configurable roles and statuses so it does not assume that one particular group always has final authority.

Access and User Roles

Anticipated system roles may include:

- Applicants who submit TA applications
- Graduate Studies and Support administrators
- Reviewers who evaluate applicants or recommend matches
- Decision-makers who approve selections or assignments
- Read-only viewers, such as approved faculty or departmental stakeholders

The final roles, permissions, approval authority, and access to internal notes will be determined with BioSci stakeholders. UCI authentication or single sign-on is preferred if technically and institutionally feasible.

TEN-WEEK MINIMUM VIABLE PRODUCT

The combined application, matching, and selection-management scope is feasible within 10 weeks if the team prioritizes a complete basic workflow.

The minimum viable product should include:

- Student application intake
- Administrative control of application periods
- Course, section, skill, and TA-position management
- Applicant search, sorting, filtering, and schedule comparison
- Basic, explainable matching suggestions
- Selection-status tracking
- Assignment recording
- Controlled user access
- Warnings for possible schedule conflicts or duplicate assignments
- Excel or CSV exports, including a final assignment roster
- Source code, testing documentation, deployment recommendations, and administrator instructions

Advanced matching, complex integrations, and formal employment processing should not displace these core capabilities.

OUT OF SCOPE

The following functions are outside the proposed 10-week minimum viable product:

- Autonomous or AI-determined final selections
- Formal hiring or employment eligibility processing
- Payroll processing
- Appointment-letter generation
- Replacement of official UCI HR or student-information systems
- Capstone-team access to live or identifiable student records
- Complex integrations requiring extended institutional approval
- Advanced optimization across every possible assignment rule

PROPOSED WORK PLAN

Weeks 1–2: Discovery and Workflow Definition

The team will interview stakeholders, document the current process, confirm the MVP, identify user roles and statuses, establish matching criteria, and define synthetic test scenarios.

Weeks 3–4: Architecture and Interface Design

The team will establish the data model, system architecture, security approach, and prototypes for the student and administrative interfaces.

Weeks 5–7: Core Development

The team will implement application intake, application-period controls, course and position management, applicant review and filtering, basic matching assistance, and selection tracking.

Weeks 8–9: Testing and Refinement

The team will test the system with synthetic records, address usability and accessibility findings, validate exports, and test schedule-conflict and duplicate-assignment warnings.

Week 10: Handoff

The team will demonstrate the completed system and deliver the source code, documentation, known limitations, and recommendations for hosting, security review, maintenance, and future development.

EXPECTED DELIVERABLES

At the end of the capstone, the student team should provide:

1. A functional web-based TA application, matching, and selection-management system.
2. A student-facing application interface.
3. An administrative interface for managing application periods, applicants, courses, sections, skills, and TA positions.
4. Applicant search, sorting, filtering, and schedule-comparison capabilities.
5. Basic, explainable student-course matching recommendations.
6. A workflow for recording recommendations, selections, assignments, and related statuses.
7. A dashboard or comparable view of proposed, filled, and unfilled TA positions.
8. Controlled access for administrators, reviewers, decision-makers, and approved viewers.
9. Excel or CSV exports, including a final assignment roster.
10. Application and database source code.
11. Testing documentation and known limitations.
12. Technical and deployment documentation.
13. A nontechnical administrator guide.
14. Recommendations for hosting, security, maintenance, and future development.
15. A final demonstration and knowledge-transfer session.

MEASURES OF SUCCESS

The project will be considered successful if the completed system:

- Creates a centralized source of applicant, course, and selection information
- Allows students to submit availability and qualifications through a clear, accessible interface
- Enables administrators to open and close application periods without developer assistance
- Allows authorized users to review and filter applicants efficiently
- Connects course requirements with applicant skills, experience, interests, and availability
- Produces understandable matching suggestions

- Preserves human control over all selection and assignment decisions
- Allows authorized users to record recommendations and final selections
- Tracks proposed, filled, and unfilled positions
- Identifies possible schedule conflicts or duplicate assignments
- Exports a usable final assignment roster
- Can be operated by nontechnical administrators with appropriate documentation

POTENTIAL CHALLENGES AND CONTINGENCY PLANS

FERPA and Student Information

TA applications may contain student information protected by FERPA. The capstone team will not receive or work with live, identifiable, or production student records.

Development, demonstrations, and testing will use synthetic test-student records representing the required fields and scenarios without corresponding to actual students. Any future testing with real student information will be conducted only by authorized UCI personnel after the appropriate privacy, security, and institutional reviews.

Assignment Governance and Approval Roles

The individuals responsible for reviewing applicants, recommending matches, approving selections, and recording assignments may vary by course or program.

The team will begin with stakeholder interviews and workflow mapping. Where feasible, the system will use configurable roles, permissions, and statuses rather than relying on one fixed approval structure.

Applicant Eligibility

It has not yet been determined which undergraduate, master's, and PhD students will be eligible to apply.

The application will require applicants to identify their student level. Detailed eligibility rules will be confirmed during discovery, and the system should accommodate future changes.

Course and Staffing Data

Course schedules, section times, staffing needs, and skill requirements may come from multiple sources or inconsistent formats.

The initial system may support manual data entry or structured spreadsheet upload. Integration with other UCI systems can be evaluated as a future enhancement.

Authentication, Hosting, and Production Access

Access to UCI authentication, hosting, or production systems may require institutional review extending beyond the capstone period.

If necessary, the team will develop and test the system in an approved development environment using test authentication and synthetic records. The team will document a recommended path to UCI authentication, hosting, security review, and production deployment.

Matching Complexity

TA assignments may involve considerations that cannot all be represented algorithmically.

The MVP will prioritize applicant filtering, schedule compatibility, skill alignment, experience, interests, and explainable matching suggestions. Advanced optimization will be treated as a stretch goal. Authorized personnel will retain control over every selection.

Limited In-House Technical Expertise

BioSci can provide operational expertise but does not have specialized software-development staff assigned to the project.

The capstone team will be expected to recommend a secure and maintainable technical approach, explain its decisions in accessible language, and provide thorough deployment, administration, and maintenance documentation.

RESOURCES BIOSCI GRADUATE STUDIES AND SUPPORT WILL PROVIDE

BioSci expects to provide:

- Regular meetings with Graduate Studies and Support stakeholders
- Operational knowledge of the current TA application and selection process
- A sample TA application form and field structure used by another UCI school
- Workflow requirements and representative application, matching, and selection scenarios
- Synthetic test-student records for development and testing
- Sample course schedules, sections, TA positions, and skill requirements, when available

- Access to representative administrators, reviewers, and decision-makers
- Stakeholder feedback throughout design, development, and testing
- Test users for usability and acceptance testing
- BioSci website guidance and coordination with appropriate UCI technical personnel when available
- Timely review of project milestones and deliverables

No live or identifiable student information will be provided to the capstone team.

PROGRAMMING LANGUAGES, SKILLS, AND OTHER REQUIREMENTS

BioSci does not require a specific programming language or framework. The team should recommend a modern, secure, accessible, and maintainable web application architecture suitable for eventual use within the UCI environment.

Helpful team capabilities include:

- Full-stack web application development
- Responsive and accessible interface design
- Relational database design
- Authentication and role-based authorization
- Secure handling of student-related information
- Search, filtering, and data export
- Schedule-conflict and duplicate-assignment detection
- Criteria-based matching and explainable recommendation design
- Workflow and status-management design
- Requirements gathering and process analysis
- Testing with synthetic data
- Deployment and hosting planning
- Technical and nontechnical documentation

The completed system should follow applicable accessibility, privacy, and security practices. Nontechnical administrators should be able to manage application periods, courses, desired skills, TA positions, and routine workflow settings.

POST-SELECTION DISCOVERY AREAS

If the project is selected, BioSci and the capstone team will confirm:

- Applicant eligibility rules

- Final application fields and validation requirements
- Recommendation, selection, and approval authority
- Administrator, reviewer, decision-maker, faculty, and viewer permissions
- Selection statuses and approval sequence
- Course, section, and staffing-data sources
- Required and preferred matching criteria
- Weighting and explanation of matching recommendations
- Treatment of schedule conflicts and multiple assignments
- Student notification requirements
- Data-retention and historical-reporting requirements
- UCI authentication, accessibility, branding, security, and hosting requirements
- Long-term ownership, maintenance, backup, and support
- Final acceptance criteria and testing responsibilities